Proof. The only difference between our MM-sr and the standard M-GT is that we ignore abstentions. To adapt the previous bounds, consider the sample s which is derived by replacing all $x = (c, \text{IDK})$ with simply $x = \text{IDK}$ for any c, but otherwise leaving x unchanged. This collapses all IDK responses into identical examples. Thus GT may count at most one extra singleton compared to sr,

$$\text{GT} - \text{sr} \in \left\{0, \frac{1}{N}\right\}.$$

The above substitution induces a distribution ϕ where $\phi(\text{IDK}) = \sum_c \mu(c)p(\text{IDK} \mid c)$ is the probability of abstaining. Similarly, we have $M - \text{MM} \in \{0, \phi(\text{IDK})\}$ with $M - \text{MM} = \phi(\text{IDK})$ if $\text{IDK} \notin s$, which happens with probability $(1-\phi(\text{IDK}))^N$. But we also have $(1-\phi(\text{IDK}))^N \leq \gamma/5$ if $\phi(\text{IDK}) \geq \frac{1}{N}\ln\frac{5}{\gamma}$. Hence, regardless of the value of $\phi(\text{IDK})$,

$$\Pr\left[M - \text{MM} \in \left[0, \frac{1}{N}\ln\frac{5}{\gamma}\right]\right] \geq 1 - \frac{\gamma}{5}.$$

Combining the above two displayed equations gives,[6]

$$\Pr\left[\left|(M - \text{GT}) - (\text{MM} - \text{sr})\right| \leq \frac{1}{N}\ln\frac{5}{\gamma}\right] \geq 1 - \frac{\gamma}{5}. \tag{7}$$

Corollary 3 at $\frac{4}{5}\gamma$ gives,

$$\Pr\left[\ |M - \text{GT}| \leq \frac{1}{N} + 2.42\sqrt{\frac{\ln(5/\gamma)}{N}}\ \right] \geq 1 - \frac{4}{5}\gamma.$$

Combining with Eq. (7) gives, by the union bound and triangle inequality,

$$\Pr\left[\ |\text{MM} - \text{sr}| \leq \frac{1}{N}\ln\frac{5}{\gamma} + \frac{1}{N} + 2.42\sqrt{\frac{\ln(5/\gamma)}{N}}\ \right] \geq 1 - \gamma.$$

Finally, the lemma follows from the fact that for $z := \frac{2}{N}\ln\frac{5}{\gamma} \geq \frac{1}{N}\ln\frac{5}{\gamma} + \frac{1}{N}$, we have $z \leq \sqrt{z}$ as long as $z \leq 1$ (otherwise the Lemma holds trivially because the bound is > 2). □

Lemma 2. *For any $N \geq 1$, $\gamma \in (0, 1]$, and any algorithm outputting $\hat{p}$,*

$$\Pr\left[2\ \text{err}_{\text{iiv}} \geq \text{sr} - \frac{6\ln(3N/\gamma)}{\sqrt{N}}\right] \geq 1 - \gamma.$$

Proof. By Lemma 1,

$$\Pr\left[\ |\text{MM} - \text{sr}| \leq 4.42\sqrt{\frac{\ln(10/\gamma)}{N}}\ \right] \geq 1 - \frac{\gamma}{2}.$$

Note that $\sqrt{\ln(10/\gamma)} \leq \ln(3N/\gamma)$ for $N \geq 2$ (and the lemma holds trivially for $N = 1$). Also, $\sqrt{2} + 4.42 \leq 6$. Hence, it suffices to show that,

$$\Pr\left[2\ \text{err}_{\text{iiv}} \geq \text{MM} - \sqrt{\frac{2}{N}}\ln\frac{3N}{\gamma}\right] \geq 1 - \frac{\gamma}{2}.$$

[6] This follows from the fact that that both $A := M - \text{MM}$ and $B := \text{GT} - \text{sr}$ are non-negative. If $0 \leq A \leq \frac{1}{N}\ln\frac{5}{\gamma}$ and $0 \leq B \leq \frac{1}{N}$, because $\frac{1}{N} \leq \frac{1}{N}\ln\frac{5}{\gamma}$, the larger of the two upper bounds is $\frac{1}{N}\ln\frac{5}{\gamma}$, so $|A - B| \leq \frac{1}{N}\ln\frac{5}{\gamma}$.

Let $\zeta := \ln(3N/\gamma)/N$ and the probability of each query appearing with an answer (not IDK) according to p to be:

$$\mu'(c) := \mu(c)\alpha_c,$$

so $\mu'(c) = p(c, a_c)$ once a_c is selected. Also note that $\mathrm{MM} = \sum_{c\in\mathcal{U}} \mu'(c)$. The lemma will thus follow from the following two inequalities:

$$\Pr\left[\forall c \in \mathcal{U}\ \mu'(c) \le \zeta\right] \ge 1 - \frac{\gamma}{3} \tag{8}$$

$$\Pr\left[2\,\mathrm{err}_{\mathrm{iiv}} \ge \mathrm{MM} - \sqrt{\frac{2}{N}}\ln\frac{3N}{\gamma} \;\middle|\; \forall c \in \mathcal{U}\ \mu'(c) \le \zeta\right] \ge 1 - \frac{\gamma}{6}. \tag{9}$$

The $\mu'(c) \le \zeta$ condition will enable us to use Hoeffding bounds. For Eq. (8), note that there are $\le 1/\zeta$ queries c with $\mu'(c) \ge \zeta$. For each of these queries, the probability $c \in \mathcal{U}$ is at most $(1-\zeta)^N$. Hence, by the union bound,

$$\Pr\left[\exists c \in \mathcal{U} :\ \mu'(c) > \zeta\right] \le \frac{1}{\zeta}(1-\zeta)^N \le \frac{1}{\zeta}e^{-\zeta N} = \frac{N}{\ln(3N/\gamma)}\frac{\gamma}{3N} \le \frac{\gamma}{3},$$

which is equivalent to Eq. (8). We now move on to establish Eq. (9).

Let the indicator $\mathrm{I}[\phi]$ to denote 1 if predicate ϕ holds and 0 otherwise. The error $\mathrm{err}_{\mathrm{iiv}}$ is at least its error summed over $c \in \mathcal{U}, r \in \mathcal{R}_c$, of course, which by definition of D is,

$$\begin{aligned}
\mathrm{err}_{\mathrm{iiv}} &\ge \frac{1}{2}\sum_{c\in\mathcal{U}} \mu(c)\alpha_c \mathrm{I}[\hat{f}(c,a_c) = -] + \frac{1}{2}\sum_{c\in\mathcal{U}} \mu(c) \sum_{r\in\mathcal{R}_c\setminus\{a_c\}} \frac{\mathrm{I}[\hat{f}(c,r) = +]}{|\mathcal{R}_c| - 1} \\
&\ge \frac{1}{2}\sum_{c\in\mathcal{U}} \mu'(c) \mathrm{I}[\hat{f}(c,a_c) = -] + \frac{1}{2}\sum_{c\in\mathcal{U}} \mu'(c) \sum_{r\in\mathcal{R}_c\setminus\{a_c\}} \frac{\mathrm{I}[\hat{f}(c,r) = +]}{|\mathcal{R}_c| - 1} \\
&= \sum_{c\in\mathcal{U}} \mu'(c)\gamma_c \text{ for } \gamma_c := \frac{1}{2}\left(\mathrm{I}[\hat{f}(c,a_c) = -] + \sum_{r\in\mathcal{R}_c\setminus\{a_c\}} \frac{\mathrm{I}[\hat{f}(c,r) = +]}{|\mathcal{R}_c| - 1}\right)
\end{aligned}$$

Thus $\mathrm{err}_{\mathrm{iiv}} \ge \sum_{c\in\mathcal{U}} \mu'(c)\gamma_c$ with γ_c define above, and it is not difficult to see that $\gamma_c \in [0,1]$. (The $\mu'(c) \le \zeta$ condition will enable us to apply Hoeffding bounds to $\sum \mu'(c)\gamma_c$.) Thus instead of Eq. (9) it suffices to show,

$$\Pr\left[2\sum_{c\in\mathcal{U}} \mu'(c)\gamma_c \ge \mathrm{MM} - \sqrt{\frac{2}{N}}\ln\frac{3N}{\gamma} \;\middle|\; \forall c \in \mathcal{U}\ \mu'(c) \le \zeta\right] \ge 1 - \frac{\gamma}{6}. \tag{10}$$

Now for the key trick: because the algorithm's output is independent of a_c for unseen $c \in \mathcal{U}$, one can equivalently imagine the a_c's being selected for unseen $c \in \mathcal{U}$ only *after* running the algorithm on the training data to select $\hat{p}$ which determines $\hat{f}$. Thus, let us suppose that c_v will later be chosen for $c \in \mathcal{U}$ but that the training data and thus $\hat{f}$ are *already fixed*.

Then, we observe that $\mathbb{E}[\gamma_c] = 1/2$ because each $r \in \mathcal{R}_c$ contributes $1/2|\mathcal{R}_c|$ to this expectation regardless of whether it is $\hat{f}(c,r) = \pm$. This gives $\mathbb{E}[\sum_c \mu'(c)\gamma_c] = \mathrm{MM}/2$ since $\mathrm{MM} = \sum_c \mu'(c)$. Finally, we can apply the Hoeffding bound to $\sum_c \mu'(c)\gamma_c$ since $\mu'(c)\gamma_c$ are independent random variables each in $[0, \mu'(c)]$. The bound depend on,

$$\sum_{c\in\mathcal{U}} (\mu'(c))^2 \le \max_{c\in\mathcal{U}} \mu'(c) \sum_{c\in\mathcal{U}} \mu'(c) \le \max_{c\in\mathcal{U}} \mu'(c) \le \zeta \text{ if } \forall c \in \mathcal{U}\ \mu'(c) \le \zeta.$$

The Hoeffding bound thus gives,

$$\Pr\left[\sum \mu'(c)\gamma_c \le \frac{\mathrm{MM}}{2} - \sqrt{\frac{\zeta \ln(6/\gamma)}{2}} \;\middle|\; \forall c \in \mathcal{U}\ \mu'(c) \le \zeta\right] \le \frac{\gamma}{6},$$

which implies Eq. (10) since $\sqrt{2\zeta \ln(6/\gamma)} = \sqrt{2\ln(3N/\gamma)\ln(6/\gamma)/N} \le \ln(3N/\gamma)\sqrt{2/N}$ (using $\ln(6/\gamma) \le \ln(3N/\gamma)$ for $N \ge 2$ and again the lemma holds trivially for $N = 1$). □

We now prove Theorem 2.

Proof of Theorem 2. The following more general lower bound, for any $\gamma \in (0, 1]$, follows directly from Theorem 1, with $\max_c |\mathcal{V}_c| = 2$, and Lemma 2. Specifically, with probability $\ge 1 - \gamma$:

$$\mathrm{err} \ge \mathrm{sr} - \frac{2}{\min_c |\mathcal{E}_c|} - \frac{6\ln(3N/\gamma)}{\sqrt{N}} - \delta.$$

For $\ge 99\%$ probability at $\gamma = 0.01$, we use the simplification that $6\ln(3N/\gamma) \le 35 + 6\ln N$. Now let $L := \max_c |\mathcal{E}_c|$.

For the upper bound, we now show that there is an efficient algorithm outputting calibrated $\hat{p}$ (so $\delta = 0$), and with probability $\ge 1 - \gamma$,

$$\mathrm{err} \le \mathrm{sr} - \frac{\mathrm{sr}}{L+1} + 5\sqrt{\frac{\ln(5/\gamma)}{N}}.$$

The 99% probability bound in the theorem follows from $5\sqrt{\ln(500)} \le 13$.

The calibrated language model learning algorithm memorizes a_c for (c, a_c) seen in the training data and agrees perfectly with p on those $c \notin \mathcal{U}$ seen in the training data. For the unseen $c \in \mathcal{U}$, it abstains with the correct probability $1 - \alpha_c$ but otherwise is uniformly random over $\mathcal{R}_c$:

$$\hat{p}(c, r) := \begin{cases} 1 - \alpha_c & \text{if } r = \mathrm{IDK} \\ \alpha_c & \text{if } c \notin \mathcal{U}, r = a_c \\ \alpha_c / |\mathcal{R}_c| & \text{if } c \in \mathcal{U}, r \in \mathcal{R}_c \\ 0 & \text{otherwise.} \end{cases}.$$

It is easy to see that, for this $\hat{p}$,

$$\mathrm{err} = \sum_{c \in \mathcal{U}} \mu(c) \frac{\alpha_c}{|\mathcal{R}_c|}(|\mathcal{R}_c| - 1) \le \sum_{c \in \mathcal{U}} \mu(c)\alpha_c \frac{L}{L+1} = \mathrm{MM}\frac{L}{L+1}.$$

Finally, by Lemma 1

$$\Pr\left[\ |\mathrm{MM} - \mathrm{sr}\,| \le 5\sqrt{\frac{\ln(5/\gamma)}{N}}\ \right] \ge 1 - \gamma.$$

These imply,

$$\Pr\left[\ \mathrm{err} \le \frac{L}{L+1}\,\mathrm{sr} + 5\sqrt{\frac{\ln(5/\gamma)}{N}}\ \right] \ge 1 - \gamma.,$$